



KENAN FIRMANSYAH

GAME & INTERACTIVE-SYSTEMS RESEARCHER

Computer Engineering graduate with **six peer-reviewed publications** on adaptive game AI, dynamic difficulty, context-aware audio, and cybersickness in serious VR games. I design and run empirical studies inside Unreal Engine and Unity testbeds, combining classical AI (Q-Learning, Fuzzy Logic, Monte Carlo) with physiological and statistical analysis. Currently researching long-term memory and personality consistency for autonomous AI agents.

EDUCATION

Bachelor of Computer Engineering, Major in Game Technology

Electronic Engineering Polytechnic Institute of Surabaya (2020 - 2024)
GPA 3.67 (Cumlaude)

Thesis: *Xanthous* — layered NPC AI for a serious VR horror game (Fuzzy Logic, Monte Carlo, FSM, Behavior Trees).

RESEARCH INTERESTS

- Adaptive systems for serious games & VR
- Game AI: agent behavior, dynamic difficulty
- Affective computing & cybersickness
- Autonomous agents with memory & personality
- Procedural & context-aware audio

METHODS & TOOLS

- Reinforcement Learning (Q-Learning), Fuzzy Logic, Monte Carlo, Behavior Trees
- Recommender systems (deep / hybrid filtering)
- Physiological signal analysis (HRV, GSR, HR)
- Statistical modeling (hierarchical regression), experimental design & user studies
- Python (NumPy, scikit-learn, PyTorch, Jupyter)
- Unreal Engine 5 & Unity as research testbeds; C++ / C#
- Data scraping & visualization (Scrapy, BeautifulSoup)

SELECTED PUBLICATIONS

Toward Adaptive VR Design: Physiological and Cognitive Predictors of Cybersickness in Serious Games

IJHCI, Taylor & Francis | 2026 | DOI: 10.1080/10447318.2026.2684064

- Multivariate cybersickness model in a UE5 fishery-management serious game (n=105). Hierarchical regression on 94 complete cases; $R^2 = 0.358$, with session duration, cognitive load, and GSR as the strongest predictors for adaptive comfort systems.

Adaptive Radio System Using MetaSounds for Serious VR Game

IEEE Access (Vol. 13) | Nov 2, 2025 | DOI: 10.1109/ACCESS.2025.3627544

- FSM-driven adaptive audio across affective states (Happy, Excited, Intense); n=30 VR user study, rated more immersive than a non-adaptive baseline.

Dynamic Level of Difficulties Using Q-Learning and Fuzzy Logic

IEEE Access (Vol. 12) | Sep 12, 2024 | DOI: 10.1109/ACCESS.2024.3457801

- Hybrid Q-Learning + Fuzzy Logic dynamic difficulty adjustment; **+35% session length, +28% player effectiveness.**

Enhancing Serious Game Experience Through In-Game Radio Using Context-Aware Recommender System

INASS IJIES | Sep 2, 2024 | DOI: 10.22266/ijies2024.1031.26

- Deep-learning hybrid recommender adapting to weather, time, and player state; **F1 up to 0.71**, outperforming traditional baselines.

RESEARCH EXPERIENCE, PROJECTS & ADDITIONAL WORK

1 RESEARCH EXPERIENCE & PROJECTS

Clone World — Personality & Memory for Autonomous AI Agents

Independent Research | Lead Researcher / Developer | Ongoing

- Life-simulation testbed where agents retain long-term memory and shift personality based on past interactions.
- Engine-agnostic design via Hexagonal Architecture; local LLMs drive dialogue and emergent story threads.

Fisherman Manager — Dynamic Difficulty Adjustment (DDA) Research

EEPIS | Lead Programmer & Designer | 2024 | Unreal Engine 5

- Built a management-sim testbed for DDA research; combined Q-Learning with Fuzzy Logic to read player behavior and adjust game-state variables in real time. Basis for the IEEE Access (2024) DDA paper.

Xanthous — AI in a Serious VR Horror Game (Undergraduate Thesis)

EEPIS | AI Programmer | 2023 Feb - 2024 July | Unreal Engine

- Designed and implemented layered NPC AI (Fuzzy Logic, Monte Carlo, FSM, Behavior Trees, AI Perception) driving agent perception and decision-making; underlying research published.

2 ADDITIONAL PUBLICATIONS

Real Time Procedural Audio for Immersive Gameplay with MetaSounds

TCCE 2025, Trends in Computational and Cognitive Engineering | 2025 | Procedural environmental audio (rain, wind, thunder) in Unreal Engine, cutting storage while keeping realistic output.

Dynamic Day and Night Cycle Impact in a Serious VR Game

ICVR 2024, International Conference on Virtual Reality | 2024 | 10-minute day/night loop; **+25% player immersion** and increased strategic depth.

3 APPLIED DATA & OTHER EXPERIENCE

Data Scientist (Intern) — PT. eBdesk (2023, 6 months) — Python pipelines for scraping, analysis, and visualization (Selenium, Gephi, Tableau).

Game Developer — Agate (2026 Feb - 2026 May) — Production game-systems work on an unannounced title.

Apple Developer Academy @ BINUS (2025 Feb - 2025 Dec) — Interdisciplinary product research: user studies, prototyping, iterative testing; CoreML and on-device LLM integration.

Unity Certified Associate: Programmer (Sep 2024 - Sep 2027) — Issued by Unity Technologies.

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